

Yousef Aboud
900 24 8250

الموضوع:

التاريخ:

Assignment 5
Modern Algebra
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Problem 1:

$$a) \alpha^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{bmatrix}$$

$$b) \beta\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 6 & 2 & 3 & 4 & 5 \end{bmatrix}$$

$$c) \alpha\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 2 & 1 & 5 & 3 & 4 \end{bmatrix}$$

Problem 3:

~~$(1\ 2\ 3\ 4\ 5)$~~

b) $1 \rightarrow 1, 2 \rightarrow 4, 3 \rightarrow 5, 4 \rightarrow 1, 5 \rightarrow 3, 6 \rightarrow 6$
Thus, $(1\ 2\ 4)(3\ 5)(6)$

Problem 6:

(a) we observe that we can write $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 5 & 4 & 6 & 3 \end{bmatrix}$ as a product of disjoint cycles. That is, $(1\ 2)(3\ 5\ 6)(4)$. By Theorem 5.2.5, the order of the permutation is $\text{lcm}(2, 3) = 6$

Problem 8:

(c) We see that $(1\ 3\ 5\ 6\ 7) = (1\ 7)(1\ 6)(1\ 5)(1\ 3)$. It can be written by product of even number of transpositions. Thus, it is an even permutation.

(d) we see,

$$1 \rightarrow 5, 5 \rightarrow 1, 2 \rightarrow 3, 3 \rightarrow 4, 4 \rightarrow 2.$$

~~That is~~ This is the product:

$$(1\ 5)(2\ 3\ 4) = (1\ 5)(4\ 3)(4\ 2)$$

which is an odd number of transpositions. Hence, the permutation is odd.

Problem 10:

~~Proof~~ In the list of all cycle types of permutations in S_8 we have $[5][3]$.

Since $\text{lcm}(5, 3) = 15$, it is obvious that any permutation that can be ~~decomposed~~ decomposed to a ~~single~~ product of 5-cycle and 3-cycle has order 15.

We know that the order of 5-cycle is 5 and the order of 3-cycle is 3, so we can rewrite each of them as an ~~even~~ product of even number of transpositions.

So, we have a permutation of order 15, which can be written as a product of even number of transpositions in S_8 . Therefore, A_8 has an element of order 15.

Problem 17:

Proof. First, we need to show the the set of even permutations in S_n is closed under ~~success~~ the operation. It is clear that if α, β are even permutations, then $\alpha\beta$ is an even permutation as it is a product of two even permutations.

second, we have to show that this set includes the identity. For simplicity, we will call the set of even permutations in S_n the set H . Now, we have to show that ~~for any permutation~~ the identity element, call it e , exists in H . Since we already know that the identity permutation is an even permutation, by lemma 5.3.4, then it is clear that the identity permutation $e \in H$.

Next, we show that H is closed under taking inverses. We know that if we have a cycle $(a \ b)$, then its inverse is $(b \ a)$. For an even permutation α , we can find α^{-1} such that α^{-1} is the product of all inverses of the transpositions of α . That is, the first transposition in α^{-1} is the inverse of the last transposition in α and so on. Hence, α^{-1} is clearly an even permutation.

Therefore, that satisfies the proof.



Problem 23

Let H be a subgroup of S_n . We pick the subset $E \subseteq H$ such that E contains the even permutations in H and the subset $O \subseteq H$ which contains the odd permutations in H . Since H is a subgroup and it contains the identity permutation, thus, E cannot be empty.

For O , we have two possibilities:

First case $\Rightarrow$ when O is the empty set, then H only contains E and it is clear that every element in H is an even permutation.

Next, if O has at least one element, call it k . We know that the product of even and odd permutations is odd. We construct the function $\gamma: E \rightarrow O$ such that $\gamma(x) = kx$, where $x \in E$. We need to show that this function is bijective. That is, we need to show that it is injective and surjective.

For injectivity, assume $\gamma(x_1) = \gamma(x_2)$, so, $kx_1 = kx_2$, by cancellation, $x_1 = x_2$ ~~and the function~~, where $x_1, x_2 \in E$. Thus, the function is injective.

For surjectivity, we pick an element a in O and call it m . Since k is odd, then k^{-1} is odd. We observe, $\gamma(k^{-1}m) = k k^{-1}m = m$. We found an image for m .

Therefore, we found a bijection between E and O , then $|E| = |O|$ and $|E| + |O| = \frac{|H|}{2}$.

Problem 25:

First, it is not closed under the operation, as if we pick α, β which are odd permutations, we know that we can write α as a product of k odd ~~per~~ transpositions and β by m odd transpositions. Then, their product $\alpha\beta$ will be a product of $k+m$ transpositions, which is an even number. So, their product $\alpha\beta$ is not in the set of odd permutations.

Second, we proved in the lecture that the identity permutation is an even permutation. Hence the set of odd permutations in S_n does not include the identity permutation.

We provided two reasons for why the set of odd permutations in S_n is not a subgroup.

Problem 43:

We pick two transpositions $\alpha = (1\ 2), \beta = (2\ 3) \in S_n$.

We observe:

$$\alpha\beta = (1\ 2)(2\ 3) = (1\ 2\ 3)$$

$$\beta\alpha = (2\ 3)(1\ 2) = (1\ 3\ 2)$$

It is clear that $\alpha\beta \neq \beta\alpha$ and the group S_n where $n \in \mathbb{Z} \wedge n \geq 3$.